

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2023/2024

MAY EXAMINATION

UCCD1213/UCCG1213 FUNDAMENTALS OF DIGITAL MEDIA TECHNOLOGY

MONDAY, 20 MAY 2024

TIME : 2.00 PM – 4.00 PM (2 HOURS)

EXAMINATION FOR BACHELOR DEGREE

Instruction to Candidates :

This question paper consists of **THREE (3)** questions in **Section A** and **TWO (2)** questions in **Section B**.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**.
Each question carries 25 marks.

Should a candidate answer more than **ONE (1)** question in **Section B**, marks will only be awarded for the **FIRST** question in that section in the order the candidate submits the answers.

Candidates are allowed to use calculator.

Answer questions only in the answer booklet provided.

UCCD1213/UCCG1213 FUNDAMENTALS OF DIGITAL MEDIA TECHNOLOGY**Section A [Total: 75 marks]**

This section consists of **THREE (3) COMPULSORY** questions.

Q1. Multimedia is the integration of multiple forms of media. The term *media* refers to text, graphics, drawings, still and moving images (video), animation, audio, and any other media where every type of information can be represented, stored, transmitted and processed digitally.

(a) List **ANY FIVE (5)** types of multimedia productions. (5 marks)

(b)



Figure 1.0 Information Kiosk

Figure 1.0 above shows an example of information kiosk, which is commonly found in public places such as shopping complexes, airports etc. The information kiosk can be considered as an interactive or non-interactive kiosk? Justify your answers. (10 marks)

(c) Briefly explain the **FOUR (4)** characteristics of a multimedia system. (10 marks)
[Total : 25 marks]

Q2.

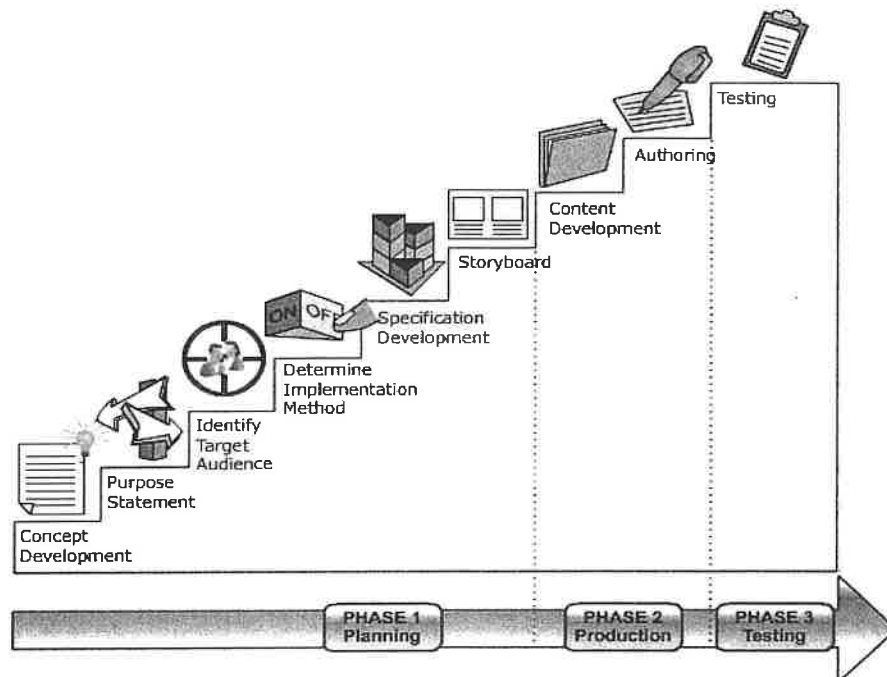


Figure 2.0 Phases in Multimedia Development Project

Figure 2.0 above shows the different stages involved in the multimedia development project. One of the stages is *storyboard*.

- (a) Briefly explain **TWO (2)** reasons why *storyboard* should always be produced before the actual development work is performed. (4 marks)
- (b) List **THREE (3)** types of storyboards. (3 marks)
- (c) Using an appropriate example/diagram, illustrate the purposes of each type of storyboard named in Q2(b). (9 marks)
- (d) In multimedia development project, elaborate **ANY THREE (3)** reasons why it is important to keep the team members well informed with the progress of the group project. (9 marks)

[Total : 25 marks]

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- Q3. (a) A text is a passage of words that conveys a set of meanings to the person who is reading it. Briefly describe the following terms: -
- (i) Kerning (2 marks)
 - (ii) Leading (2 marks)
 - (iii) Anti-aliasing (2 marks)
- (b) Provide **TWO (2)** factors that determine the storage required to store an image. (4 marks)
- (c) Given an image of any format, explain **ANY TWO (2)** factors that could determine the extent how this image can be compressed. (8 marks)
- (d) Using an appropriate example, explain your understanding of the following statement: -

“In vector graphics, an image created remains separate from others”

(7 marks)

[Total : 25 marks]

UCCD1213/UCCG1213 FUNDAMENTALS OF DIGITAL MEDIA TECHNOLOGY**Section B [Total: 25 marks]**

This section consists of **TWO (2) OPTIONAL** questions. Answer **ANY ONE (1)** question.

- Q4. (a) MIDI is a standard protocol for the interchange of musical information between musical instruments, synthesizers, and computers.
- (i) Unlike regular audio files like MP3s or WAVs, MIDI files are generally smaller in size. Justify. (4 marks)
 - (ii) Explain why MIDI files are device dependent. (4 marks)
 - (iii) Elaborate why MIDI files are not classified as digital sound. (4 marks)
 - (iv) Calculate the number of bytes required to store 2 minutes 18 seconds of stereo sound at CD quality with a resolution of 16 bits.
NOTE: Sampling frequency of CD is 44.1 kHz. (4 marks)
- (b) Digital video is moving images that have been captured and created electronically by a computer. Videos are generally huge in size. Illustrate **THREE (3)** different ways of reducing video file size to enable more efficient transfer through low bandwidth networks. (9 marks)
- [Total : 25 marks]

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Q5. (a) Briefly explain on how to design good user interface. (7 marks)

(b)

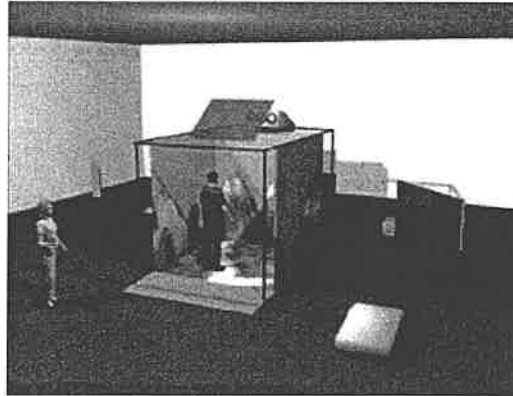


Figure 3.0 Cave Automatic Virtual Environment

Figure 3.0 shows Virtual Reality (VR) environment consisting of a cube-shaped VR room or a room-scale area in which the walls, floors and ceilings are projection screens.

(i) Briefly explain **TWO (2)** advantages of CAVE as compared to Head-mounted Display (HMD). (6 marks)

(ii) Is CAVE (Cave Automatic Virtual Environment) classified as immersive or non-immersive VR? Explain your answer. (6 marks)

(c) Augmented Reality (AR) has many different implementation models and applications, but its primary objective is to provide a rich audiovisual experience. AR works by employing computerized simulation and techniques such as image and speech recognition, animation, head-mounted and hand-held devices and powered display environments to add a virtual display on top of real images and surroundings. Elaborate **TWO (2)** differences between Virtual Reality and Augmented Reality. (6 marks)

[Total : 25 marks]